Stability without concealment

When does grouping correlated inputs improve explanation stability without hiding useful information?

THE PROBLEM

Similar predictions can produce different feature rankings after retraining.

THE TRAP

Shorter lists and random grouping can also appear more stable.

OUR TEST

Compare matched groups; measure stability plus fidelity and cancellation.

Credit benchmark • interpretable baseline • controlled comparisons • eight weeks

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What the literature establishes — and what we will test

SHAP + dependence	Credit depends on model, background and explanation game. Lundberg & Lee; Aas et al.
Disagreement + aggregation	Agreement is a research problem; fusion already exists. Krishna et al.; Bhatt et al.
Grouped explanations	groupShapley and recent robustness work are close prior art. Jullum et al.; Hendro et al. (2025)

Proposed gap: test stability beyond matched random grouping, together with information loss and known-ground-truth fidelity.

Sources: SHAP [S01]; disagreement [S03]; groupShapley [S04]; ITNAC 2025 [N03]; agreement 2026 [N04]

Methodology and controlled evaluation

01

Split + preprocess

Fit only on training data

02

Retrain + explain

Same held-out people

03

Group + controls

Match group sizes exactly

04

Evaluate

Stability AND retained detail

MEASURE

RBO stability • matched-random difference
Cancellation • synthetic attribution error
AP / ROC-AUC / Brier • runtime

CONTROL

No grouping • same-size random groups
Known additive + interaction cases
Fixed model/output/background per comparison

Minimum scope: synthetic data + UCI credit; logistic + one tree family. HELOC is conditional on verified access.

Pilot results and eight-week validation plan

11 groups; matched group sizes



Controlled RBO difference: +0.061

30,000 rows • 24k train / 6k test
AP 0.489 • AUC 0.713 • Brier 0.148
9 checks pass; deterministic replay matches

EIGHT-WEEK VALIDATION

W1–2 Data + synthetic oracle
W3–4 Tree SHAP + matched controls
W5–6 Fidelity + replication
W7–8 Paper + advisor review

Limit: one split, logistic model only. No significance, causal recourse or publication claim.

The concepts you need before the experiment

MODEL

Inputs → learned function → default probability. Testing asks whether predictions generalize.

ATTRIBUTION

A contribution to a stated model output, relative to a specified background.

STABILITY

Does the explanation change when the model is retrained on resampled data?

FIDELITY

Does the explanation reflect the specified model/game? Stability alone is insufficient.

The “corrected” manual still needs correction

- Lower correlation cutoff → more merging, not more groups.
- Coalition sums are not generally Owen / exact group values.
- Mean logit transformed by sigmoid is not mean probability.
- Margin code has an order-statistic error; alpha direction is reversed.
- Static credit data cannot validate temporal, actionable recourse.
- Agreement, monotonicity and lower VIF do not prove explanation quality.

Detailed 36-point audit with printed PDF page numbers is in the Research Dossier.

A linear attribution you can calculate by hand

Score = $3x_1 + 2x_2$ Reference = (1, 2) Input = (2, 5)

BASE SCORE

$3\times1 + 2\times2 = 7$

CONTRIBUTIONS

$3\times(2-1) = 3$
$2\times(5-2) = 6$

RECONSTRUCTION

$7 + 3 + 6 = 16$

Always state the explained output scale. The pilot explains log-odds, not percentage points.

Illustrative arithmetic, not a measured applicant. Pilot uses the same identity on standardized logit inputs.

Stable summaries can conceal different things

CANCELLATION

Feature credits: +5 and −4.8
Net group contribution: 0.2
Unsigned internal mass: 9.8
A stable net can hide strong components.

GROUPING CHANGES THE GAME

Three-player unanimity game:
ordinary A + B = 2/3
merged {A,B} group = 1/2
Coalition sum ≠ exact group value.

Retain group summaries AND feature drill-down; test interactions with a small exact oracle.

Coalition sum versus grouped game: groupShapley [S04]. Worked numbers are illustrative.

Pilot provenance and what its numbers mean

DATA + MODEL

30,000 observed customer records
19 numeric inputs after exclusions
24,000 train / 6,000 test
500 held-out audit rows
10 bootstrap logistic models
Exact interventional logit attribution

MEASUREMENTS

19-feature RBO: 0.776
11-group RBO: 0.924
Size-matched random RBO: 0.863
Controlled difference: +0.061
Max reconstruction error: 2.22e−15
No inferential p-value claimed

Single split + simple model + fixed partition. Fidelity, second dataset and full benchmark remain unrun.

Full experiment: isolate one source of change at a time

- Data: controlled synthetic cases + UCI; verified HELOC if available.
- Models: logistic regression first; one small tree ensemble next.
- Split: 60% train / 20% development / 20% locked test.
- Separate: model resampling, explainer sampling and reclustering.
- Controls: singleton, random same-size, domain and giant groups.
- Small exact group games verify additive and interaction cases.

Evaluation must measure more than consistency

Stability	RBO at p=.85; sensitivity at .7/.95; fixed audit individuals.
Controlled gain	Correlation-group stability minus size-matched random-group stability.
Retained information	Cancellation, drill-down, exact synthetic error and masker sensitivity.
Model + uncertainty	AP / AUC / Brier; paired effects; honest conditional uncertainty.

What would falsify the project hypothesis?

EVIDENCE AGAINST

Random groups perform equally well.
Information loss offsets stability.
Results disappear on new splits.
Closest prior art already answers the question.

HONEST RESPONSE

Report the failure conditions.
Keep all runs and controls.
Revise claims and scope.
Do not invent expected results or tune on test.

Research quality comes from a valid answer, not from forcing a positive outcome.

The code demonstration: two minutes, no hand-waving

SHOW THE FILES

demo.py — measured baseline
test_demo.py — 9 checks
download_data.py — official input
results/pilot.json — traceable evidence
Colab_Quickstart.ipynb — CPU route

EXPLAIN THE CHECKS

Disjoint training/test IDs
Complete-linkage condition
Hand-computed RBO and linear values
Exact logit reconstruction
Identical deterministic rerun

Not implemented yet: tree SHAP, LIME, full synthetic oracle, recourse, second-data validation.

Eight weeks, with a deliverable gate each fortnight

WEEKS 1–2	Understand baseline; audit data; verify exact synthetic games.
WEEKS 3–4	Add tree SHAP; run matched-group stability and negative controls.
WEEKS 5–6	Measure information loss; replicate; freeze main results.
WEEKS 7–8	Write, review, replay and prepare a venue-specific submission.

Publication plan: target fit, not a promise

PRIMARY FIT

Future ACM ICAIF call
Financial ML + explainability
2026 submissions closed
Verify next CFP with advisor

EVALUATION OPTION

ECML PKDD XAI / trustworthy-ML workshop
Check actual workshop and archival policy

STRETCH

PAKDD research track
Requires stronger methodological evidence
No deadline verified

Ask the supervisor: Is the narrowed empirical study suitable, and what publication format does DTU require?

Selected references and the novelty checkpoint

- Lundberg & Lee (2017). A Unified Approach to Interpreting Model Predictions. arXiv:1705.07874.
- Krishna et al. The Disagreement Problem in Explainable Machine Learning. arXiv:2202.01602.
- Jullum, Redelmeier & Aas. groupShapley. arXiv:2106.12228.
- Chen et al. (2020). True to the Model or True to the Data? arXiv:2006.16234.
- Hendro, Mei & Welch (2025). Feature-Grouping for Robust SHAP. DOI: 10.1109/ITNAC66378.2025.11302604.
- Quantifying overall agreement… (2026). DOI: 10.1007/s12065-026-01145-y.